

Cylinder Inscribed in a Cone Optimization Problem

Determine the dimensions of the right circular cylinder of maximum volume that can be inscribed in a cone having a diameter of 40cm and a height of 30cm.

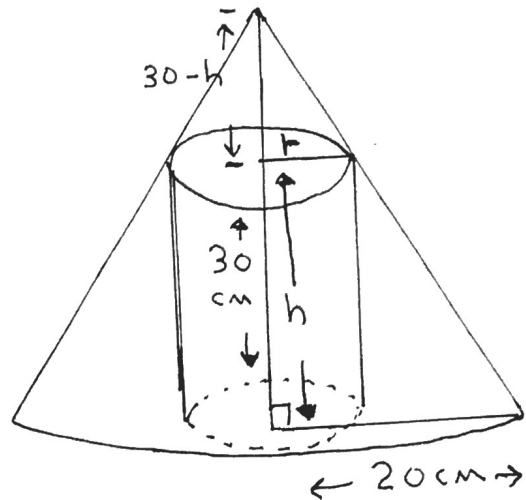
$$V = \pi r^2 h \quad (1)$$

By similar triangles,

$$\frac{30-h}{30} = \frac{r}{20}$$

$$30r = 20(30-h)$$

$$r = \frac{2(30-h)}{3} \quad (2)$$



Substitute (2) into (1)

$$V = \pi \left[\frac{2(30-h)}{3} \right]^2 \cdot h$$

$$V = \frac{4}{9} \pi h (30-h)^2$$

$$\text{Now, } V' = \frac{4}{9} \pi \left[(30-h)^2 + h \cdot 2(30-h)(-1) \right]$$

$$V' = \frac{4}{9} \pi \left[(30-h)^2 - 2h(30-h) \right]$$

$$V' = \frac{4}{9} \pi (30-h)(30-h-2h)$$

$$V' = \frac{4}{9} \pi (30-h)(30-3h)$$

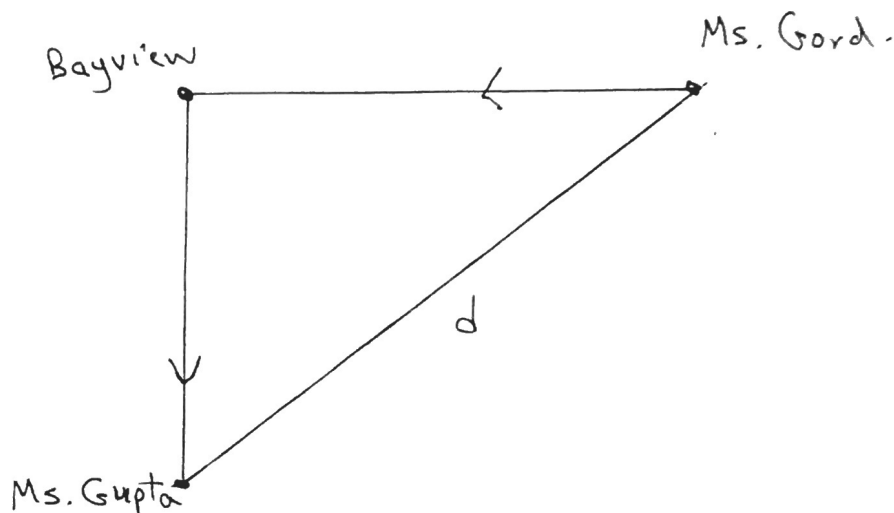
$$V' = 0 \Rightarrow h = \underset{\substack{\uparrow \\ \text{inadmissible}}}{30} \text{ or } h = 10 \Rightarrow r = \frac{40}{3}$$

$\therefore$ The height is 10 cm and the radius is $\frac{40}{3}$ cm

Closest Together Optimization Problem

Ms. Gupta leaves Bayview Secondary School at 1:00pm and travels due south at 80 km/hr. At 1:00pm, Ms. Gord is heading due west at 90 km/hr and arrives at Bayview Secondary School at 2:00pm. Determine the approximate time of day (hour and minute) at which Ms. Gupta and Ms. Gord were **closest** together.

$$d = s \cdot t$$



* Ms. Gupta's distance from Bayview equals $80t$.

* Ms. Gord's distance from Bayview equals $90(1-t)$. $[90 - 90t]$

$$d = \sqrt{(80t)^2 + [90(1-t)]^2}$$

$$d' = \frac{2(80t) \cdot 80 + 2[90(1-t)](-90)}{2\sqrt{(80t)^2 + [90(1-t)]^2}}$$

$$d' = 0 \Rightarrow 6400t - 8100 + 8100t = 0$$

$$14500t = 8100$$

$$t = \frac{8100}{14500}$$

$$t \approx 0.559 \text{ hrs } (\text{OR}) \sim 1:34 \text{ pm}$$